ENVIRONMENTAL SCIENCE

BHM4T02 · NEP 2020 · RTMNU / GNIET

B.Tech CSE (Data Science) · Semester IV

UNIT 1: AIR POLLUTION AND ITS CONTROL TECHNIQUES

■ **Total Credits 02 T**	■ **University Exam 70 Marks**	■ **College Assessment 30 Marks**	■ **Lectures 6 Hours (Unit 1)**

■ SYLLABUS COVERAGE

✔ Contaminant behaviour in the environment

✔ Air pollution due to SOx, NOx, Photochemical Smog, Indoor Air Pollution

✔ Natural pathways for degradation: Carbon Cycle, Sulphur Cycle, Nitrogen Cycle, Oxygen Cycle

✔ Factors altering atmosphere: Deforestation, Fossil Fuels, Industrial & Vehicular Emissions, CFCs

✔ Techniques to control Air Pollution (Cyclone, ESP, Bag Filter, Wet Scrubber, Catalytic Converter)

✔ Ambient Air Quality & Continuous Air Quality Monitoring (NAAQS, AQI, CAAQM)

✔ Control measures at source, Kyoto Protocol & Carbon Credits

Includes RTMNU 7-Mark OR-Type Questions with Answer Guides

SECTION 1: CONTAMINANT BEHAVIOUR IN THE ENVIRONMENT

1.1 Definition of a Contaminant

A **contaminant** is any physical, chemical, biological, or radiological substance present in the environment — air, water, or soil — at concentrations sufficient to cause harm to human health, living organisms, or the overall ecosystem. Contaminants differ from pollutants in that a contaminant is simply a substance out of place, while a pollutant is a contaminant that actively causes measurable damage.

1.2 Sources of Contaminants

■ A. Industrial & Commercial Facilities

Factories, chemical plants, refineries, and mining operations release toxic wastes — heavy metals (lead, mercury, cadmium), acids, solvents — into air, water, and soil through exhaust stacks, effluent channels, and improper dumping. These are classified as **point sources** because pollution originates from a single identifiable location.

■ B. Oil & Chemical Spills

Accidental spills from oil tankers, pipelines, or offshore drilling platforms contaminate oceans, coastlines, and groundwater with petroleum hydrocarbons (benzene, toluene, xylene) and toxic chemicals. These are persistent organic pollutants (POPs) that resist natural degradation.

■ C. Non-Point Source Pollution (NPS)

Non-Point Source Pollution is contamination that **cannot be traced to a single discharge point**. Stormwater runoff from agricultural fields, roads, and parking lots carries fertilisers, pesticides, and oils into water bodies. NPS is the leading global cause of water quality degradation and is extremely difficult to regulate.

> ■ **NOTE:** NPS vs Point Source: Point source = single identifiable outlet (factory pipe, sewage drain). NPS = diffuse, spread over large areas (agricultural runoff, urban stormwater). NPS is harder to control because it has no single origin.

■ D. Sewage Systems & Wastewater Plants

Improperly treated municipal sewage discharges pathogens (bacteria, viruses), heavy metals, and high BOD organic matter into rivers and coastal waters. In India, approximately **80% of sewage is discharged untreated**.

■ E. Hazardous Waste Sites

Old industrial sites, landfills, and illegal dumping zones leach persistent contaminants — heavy metals, chlorinated solvents, radioactive materials — into groundwater and surrounding soil for decades.

1.3 Key Contaminant Behaviour Processes

■ Bioaccumulation

Bioaccumulation is the gradual build-up of a contaminant in an organism's body at a rate faster than it can be eliminated. Persistent organic pollutants (POPs) like DDT and heavy metals like mercury dissolve in fatty tissues and accumulate over an organism's lifetime. A fish in mercury-contaminated water accumulates far more mercury in its body than the surrounding water contains.

■ Biomagnification

Biomagnification is the progressive *increase* in concentration of a contaminant at each successive trophic level of the food chain. A small amount of DDT in phytoplankton becomes higher in zooplankton, higher still in small fish, and highest in apex predators (eagles, humans) — often **thousands of times higher** than the original environmental concentration. This is why predatory fish like tuna and swordfish carry dangerous mercury levels.

> ■ **NOTE:** DDT was banned in India in 2020. Mercury from coal power plants enters oceans → bioaccumulates in phytoplankton → biomagnifies through the food chain → reaches dangerous levels in tuna and swordfish consumed by humans.

> ■ **EXAM TIP:** 7-Mark Q: Explain contaminant behaviour in the environment. Include sources, bioaccumulation, and biomagnification with examples.

SECTION 2: AIR POLLUTION — DEFINITION & CLASSIFICATION

2.1 Definition of Air Pollution

Air pollution is the presence of one or more harmful substances — gases, particulates, or biological molecules — in the atmosphere in concentrations sufficient to cause harm to human health, living organisms, the natural environment, or the built environment. It can originate from **natural sources** (volcanic eruptions, dust storms, wildfires) or **anthropogenic sources** (industries, vehicles, burning of fossil fuels).

2.2 A. Primary Pollutants — Emitted Directly from a Source

- **Carbon Monoxide (CO):** Produced by incomplete combustion of fuel in vehicles and industries. Colourless and odourless — the 'silent killer.' Binds to haemoglobin 240x more strongly than oxygen → carboxyhaemoglobin → reduces oxygen delivery to tissues.
- **Sulfur Dioxide (SO_2):** Released by burning coal and oil in power plants, and volcanic eruptions. Irritates airways; reacts with atmospheric moisture to form sulfuric acid (H_2SO_4) — the main cause of acid rain.
- **Nitrogen Oxides (NO_x):** NO and NO_2 from high-temperature combustion in vehicle engines and thermal power plants. Precursors to photochemical smog and acid rain.
- **Particulate Matter ($PM_{2.5}$ / PM_{10}):** Microscopic dust, smoke, and exhaust particles. $PM_{2.5}$ (< 2.5 µm) penetrates deep into alveoli, entering the bloodstream — causes cardiovascular disease and lung cancer.
- **Volatile Organic Compounds (VOCs):** Unburned fuel vapours from vehicles and industry. React with NO_x in sunlight to form photochemical smog.
- **Lead (Pb):** From battery industries and smelters. A neurotoxin causing permanent cognitive impairment in children.

2.3 B. Secondary Pollutants — Formed by Atmospheric Reactions

- **Ground-Level Ozone (O_3):** Formed when NO_x and VOCs react in the presence of UV sunlight. Beneficial in stratosphere; toxic at ground level — damages lungs and crops.
- **Acid Rain (H_2SO_4 / HNO_3):** SO_2 / NO_x + H_2O → sulfuric / nitric acid → dissolves in rain → pH < 5.6. Damages lakes, forests, and stone buildings.
- **PAN (Peroxyacetyl Nitrate):** Formed from ozone + VOCs. A key component of photochemical smog. Causes severe eye irritation and plant damage — 10x more toxic to plants than ozone.

■ **EXAM TIP:** 7-Mark Q: Classify air pollutants into primary and secondary. Explain any three primary pollutants in detail.

SECTION 3: IMPACT OF SOx (SULFUR OXIDES)

3.1 Sources of SOx

- Burning of coal (1–4% sulfur content) and petroleum in thermal power plants and industries — ~50% of global SO_2 emissions
- Volcanic eruptions — major natural source (Mt. Pinatubo, 1991: 20 million tonnes SO_2 in one eruption)
- Metal smelting — roasting of sulfide ores (FeS_2, CuS) releases SO_2
- Petroleum refining and diesel vehicle exhaust

3.2 Health Effects of SOx

■ Respiratory Effects

SO_2 is a highly soluble, acidic gas that dissolves in the moisture lining the respiratory tract, producing sulfurous acid. This irritates mucous membranes, causing **coughing, wheezing, and shortness of breath**. Even brief exposure at 0.5 ppm worsens pre-existing asthma. Prolonged exposure leads to **Chronic Obstructive Pulmonary Disease (COPD)** — an irreversible, progressive lung disease.

■ Eye & Mucous Membrane Irritation

SO_2 causes burning sensations in the eyes, lacrimation (tearing), conjunctivitis, and irritation of nasal passages. Workers in sulfur-rich industrial environments face constant exposure.

■ Cardiovascular Effects

Chronic SO_2 exposure triggers systemic inflammation and oxidative stress, increasing the risk of cardiovascular disease. Fine sulfate $PM_{2.5}$ particles formed from SO_2 reactions further contribute to cardiovascular mortality.

3.3 Environmental Effects of SOx

■ Acid Rain Formation

SO_2 reacts with atmospheric water vapour and oxygen to form sulfuric acid, which dissolves in rainfall producing acid rain (pH below 5.6). Acid rain: **acidifies lakes** killing fish and aquatic life; **leaches calcium and magnesium** from forest soils causing forest dieback; **corrodes stone buildings** through: $H_2SO_4 + CaCO_3 \rightarrow CaSO_4 + CO_2 + H_2O$.

■ Sulfate Aerosols

SO_2 in the atmosphere reacts to form fine sulfate aerosol particles ($PM_{2.5}$). These aerosols reduce visibility, contribute to lung disease, and have a temporary cooling effect on climate by reflecting incoming sunlight (global dimming).

3.4 Control of SOx Pollution

- **Fuel Switching:** Replace high-sulfur coal with natural gas or renewable energy
- **Flue Gas Desulphurisation (FGD):** Wet scrubbers spray lime slurry ($Ca(OH)_2$) to absorb SO_2 → converts to gypsum ($CaSO_4$). Removes up to 95% of SO_2 from power plant exhaust
- **Coal Desulphurisation:** Physical and chemical treatment of coal before combustion to remove sulfur compounds
- **Low-Sulfur Fuels:** Use low-sulfur diesel; MARPOL Annex VI limits sulfur in marine fuel to 0.5%

■ **EXAM TIP:** 7-Mark Q: Explain sources and impact of SOx on human health and environment. Describe control measures including FGD.

SECTION 4: IMPACT OF NOx (NITROGEN OXIDES)

4.1 Sources of NOx

- Vehicle engines (petrol and diesel) — the largest urban source of NOx
- Thermal power plants burning coal, gas, or oil
- Industrial furnaces, cement kilns, and steel plants
- Lightning — a natural source producing NO from atmospheric N2 and O2
- Aircraft engines at high altitude

4.2 Health Effects of NOx

■ Airway Irritation & Lung Inflammation

NO2 is a reddish-brown gas with a pungent smell. It inflames the lining of the lungs, causing persistent coughing, wheezing, and chest tightness. Unlike SO2 which affects upper airways, NO2 penetrates **deeper into the alveoli**, causing oxidative damage to lung tissue.

■ Aggravation of Pre-existing Conditions

NO2 worsens asthma, bronchitis, emphysema, and COPD. Children exposed to high NOx levels show measurably **stunted lung development**, permanently reducing lung capacity and increasing susceptibility to respiratory infections.

■ Photochemical Smog

NOx — especially NO2 — is the key precursor to photochemical smog. In sunlight, NO2 undergoes photolysis releasing atomic oxygen which forms ground-level ozone. (Detailed in Section 5.)

■ Acid Rain (HNO3)

NOx + H2O → HNO3 (nitric acid) → acid rain. NOx is responsible for approximately **30% of acid rain** in most industrialised countries.

4.3 Comparison: SOx vs NOx

Parameter	SOx (Sulfur Oxides)	NOx (Nitrogen Oxides)
Primary Gas	Sulfur Dioxide (SO_2)	Nitrogen Dioxide (NO_2)
Main Sources	Coal burning, volcanic eruptions, smelting	Vehicles, power plants, lightning
Health Effect	Coughing, asthma, COPD	Airway inflammation, lung damage
Eye Effects	Burning sensation, conjunctivitis	Minimal direct eye irritation
Secondary Pollutant	Acid rain (H_2SO_4), sulfate PM2.5	Photochemical smog, ozone, HNO_3
Gas Colour	Colourless (sharp smell)	Reddish-brown (pungent)

4.4 Control of NOx Pollution

- **Three-Way Catalytic Converters:** Platinum/palladium/rhodium catalysts convert NOx → N_2, CO → CO_2, HC → CO_2+H_2O. Efficiency ~90%
- **Selective Catalytic Reduction (SCR):** At power plants, ammonia injected into flue gas reacts over a catalyst: NOx + NH_3 → N_2 + H_2O
- **BS-VI Emission Norms:** India adopted Bharat Stage VI (equivalent to Euro 6) from April 2020 — reduces NOx from diesel vehicles by 70%
- **Low-NOx Burners:** Staged combustion and flue gas recirculation reduce peak flame temperatures, preventing thermal NOx formation
- **Electric Vehicles:** Zero tailpipe emissions — eliminates vehicle NOx contribution entirely

SECTION 5: PHOTOCHEMICAL SMOG

5.1 Definition

Photochemical smog (also called **'Los Angeles smog'**) is a type of **secondary air pollution** formed when nitrogen oxides (NOx) and volatile organic compounds (VOCs) from vehicle exhaust react in the presence of **UV sunlight** to produce ozone (O_3), PAN (Peroxyacetyl Nitrate), and other oxidants. It appears as a brownish-yellow haze over cities — especially in sunny, traffic-heavy urban areas with poor air circulation.

5.2 Step-by-Step Formation Mechanism

The formation occurs through a photochemical chain reaction:

- Step 1 — Emission: Vehicles and industries emit NO_2 and VOCs (unburned hydrocarbons) into atmosphere during morning rush hours
- Step 2 — Photolysis of NO_2: UV radiation from sunlight breaks NO_2: NO_2 + UV light → NO + O• (free atomic oxygen)
- Step 3 — Ozone Formation: Free atomic oxygen reacts with molecular oxygen: O• + O_2 → O_3 (ozone)
- Step 4 — Reaction with VOCs: Ozone reacts with VOCs → forms PAN, formaldehyde, aldehydes, and other oxidants
- Step 5 — Smog: The mixture of O_3 + PAN + aldehydes creates the brownish photochemical smog visible over cities by midday

5.3 Conditions Favouring Formation

- **Intense Sunshine:** High UV radiation intensity drives photochemical reactions — worst on clear, sunny days
- **Dense Vehicle Traffic:** Primary source of NOx and VOC emissions — smog peaks 2–4 hours after morning rush hour
- **Low Wind Speed:** Stagnant air allows pollutants to accumulate — calm days are worst
- **Geographical Trapping:** Mountains surrounding cities (Los Angeles, Delhi, Mexico City) trap polluted air in valleys
- **Temperature Inversion:** A layer of warm air traps cooler, polluted air below — prevents normal upward dispersal of pollutants

5.4 Effects of Photochemical Smog

■ Human Health Effects

- **Eye & Throat Irritation:** Ozone and PAN cause intense burning of eyes, lacrimation, and throat inflammation. PAN is 10x more irritating than ozone to eyes
- **Respiratory Damage:** Ozone at 0.1 ppm causes chest pain, coughing, shortness of breath. Long-term exposure permanently reduces lung function
- **Aggravation of Conditions:** Worsens asthma, COPD, bronchitis — can trigger life-threatening attacks in sensitive individuals

■ Environmental & Material Effects

- **Crop Damage (Phytotoxicity):** Ozone enters plant stomata, disrupts photosynthesis, reduces crop yields by 5–15%. Most sensitive: wheat, rice, soybeans
- **Reduced Visibility:** The brownish haze scatters light, significantly reducing visibility in urban areas
- **Material Damage:** Ozone causes rubber cracking, degrades paints, textiles, and plastic materials

5.5 Control Measures

- **Electric Vehicles (EVs):** Zero tailpipe emissions — eliminates both NOx and VOC at source
- **Catalytic Converters:** Reduce NOx and VOC from petrol/diesel vehicles by ~90%
- **CNG & LPG Adoption:** Far less NOx and VOC than petrol/diesel (e.g., Delhi's buses converted to CNG in 2001)
- **Strict Emission Norms (BS-VI):** Significantly reduced per-vehicle NOx and PM since April 2020
- **Public Transport:** Metro, buses, cycling lanes reduce private vehicle numbers — reducing NOx and VOC at source

> ■ **EXAM TIP:** 7-Mark Q: Explain the formation mechanism, conditions favouring formation, effects, and control of photochemical smog.

SECTION 6: INDOOR AIR POLLUTION (IAP)

6.1 Definition & Significance

Indoor Air Pollution (IAP) is the contamination of air within and around buildings due to physical, chemical, and biological pollutants from indoor activities, building materials, furnishings, and inadequate ventilation. Humans spend approximately **90% of their time indoors**. WHO estimates IAP causes **3.8 million deaths per year** globally. Indoor concentrations can be **2–5 times higher** than outdoor levels in modern tightly-sealed buildings.

6.2 Major Indoor Air Pollutants & Their Sources

■ Carbon Monoxide (CO)

Produced by incomplete combustion in gas stoves, kerosene heaters, and poorly maintained fireplaces. Colourless and odourless — the 'silent killer.' CO binds to haemoglobin, preventing oxygen delivery. At 200 ppm → headache and dizziness; at 800 ppm → unconsciousness within 2 hours; at 1600 ppm → death. Carbon monoxide detectors are essential.

■ Volatile Organic Compounds (VOCs)

Released ('off-gassed') from paints, varnishes, adhesives, cleaning products, synthetic carpets, and new furniture. Key VOCs: **benzene** (carcinogenic — from tobacco smoke, stored fuel), **formaldehyde** (irritant and carcinogen from pressed wood products). Cause eye/nose/throat irritation, headaches, and long-term liver and kidney damage.

■ Particulate Matter (PM2.5)

From cooking smoke (especially biomass/chulha stoves — a major problem in rural India), tobacco smoke, incense, and candles. Fine PM2.5 particles penetrate deep into alveoli and enter the bloodstream, causing cardiovascular disease and lung cancer. Cooking on traditional chulhas is a leading cause of IAP mortality in Indian rural women.

■ Biological Pollutants

Include mold and mildew spores (damp walls/bathrooms), dust mites (carpets, mattresses), pet dander, cockroach allergens, and bacteria/viruses. These trigger allergies, asthma attacks, and respiratory infections in sensitive individuals.

■ Radon Gas

A naturally occurring radioactive gas from uranium decay in soil, seeping through basement floors and walls. Colourless and odourless. It is the **second leading cause of lung cancer globally** (after cigarette smoking). Radioactive decay products irradiate lung tissue over years of exposure. High in regions with granite-rich geology.

■ Asbestos Fibres

Formerly used in building insulation, ceiling tiles, and pipe lagging. When disturbed during renovation, microscopic fibres are released. Chronic inhalation causes **asbestosis** (lung fibrosis) and **mesothelioma** (cancer of the pleural lining of lungs). Banned in most developed countries but still present in many older Indian buildings.

6.3 Building-Related Health Syndromes

Syndrome	Description	Cause	Resolution
Sick Building Syndrome (SBS)	Headaches, fatigue, dizziness, eye irritation — no specific illness identified	Poor ventilation, VOC off-gassing, chemical pollutants	Symptoms resolve when leaving the building
Building-Related Illness (BRI)	Diagnosable illness (e.g., Legionnaires' disease — a severe pneumonia)	Specific pathogen — Legionella bacteria in HVAC cooling towers	Symptoms persist even after leaving the building

6.4 Control of Indoor Air Pollution

- **Adequate Ventilation:** Fresh air through windows, exhaust fans, and HVAC systems with HEPA filters removes and dilutes pollutants
- **Clean Cooking Fuels:** Replace biomass chulhas with LPG, CNG, or electric induction cooktops (India's Ujjwala Scheme)
- **Low-VOC Materials:** Use low-VOC paints and adhesives; allow new furniture to off-gas outdoors before bringing inside
- **Radon Mitigation:** Sub-slab depressurisation systems draw radon from beneath foundations and vent it safely outdoors
- **Ban Indoor Smoking:** Tobacco smoke is the single largest controllable source of indoor PM2.5, CO, and carcinogenic VOCs

■ **EXAM TIP:** 7-Mark Q: Explain Indoor Air Pollution — sources, pollutants (CO, VOCs, PM, Radon, Asbestos), health effects, SBS vs BRI, and control measures.

SECTION 7: FACTORS ALTERING COMPOSITION OF THE ATMOSPHERE

7.1 Normal Atmospheric Composition

The Earth's atmosphere: N_2 (78%), O_2 (21%), Ar (0.93%), CO_2 (0.04%), and trace gases. Human activities are significantly altering this balance, driving climate change, ozone depletion, and increased air pollution.

7.2 Deforestation

Forests are the 'lungs of the Earth.' Through photosynthesis, trees absorb CO_2 and release O_2, acting as critical **carbon sinks**. The Amazon rainforest alone stores approximately 150–200 billion tonnes of carbon.

■ How Deforestation Alters the Atmosphere

- Eliminates carbon sinks — atmospheric CO_2 levels rise
- Release of stored carbon — burning trees releases centuries of stored carbon as CO_2 and CH_4
- Burning of cleared forests directly releases CO_2, CO, black carbon (soot), and PM
- Deforestation contributes ~10–15% of global greenhouse gas emissions annually
- Reduces O_2 production and disrupts regional rainfall patterns through reduced evapotranspiration

7.3 Burning of Fossil Fuels

The combustion of coal, petroleum, and natural gas is the **single largest driver of human-caused climate change**. Since the Industrial Revolution (~1760), atmospheric CO_2 has risen from ~280 ppm to over **420 ppm (2024)**. Fossil fuel combustion releases approximately **36 billion tonnes of CO_2 annually**.

■ Greenhouse Effect

CO_2, methane (CH_4), water vapour, and nitrous oxide (N_2O) are greenhouse gases that absorb outgoing infrared radiation from Earth's surface and re-emit it, warming the planet. The **enhanced greenhouse effect** due to massive increases in GHG from fossil fuels drives dangerous climate change. Global average temperature has already risen by **~1.1°C above pre-industrial levels**.

■ Pollutants Released

- **CO_2:** Primary greenhouse gas driving global warming and ocean acidification
- **SO_2:** Causes acid rain and sulfate aerosols
- **NO_x:** Precursor to photochemical smog and acid rain
- **Black Carbon (Soot):** A potent short-lived climate forcer — absorbs solar radiation and warms atmosphere
- **Mercury (Hg):** Released from coal combustion — bioaccumulates in aquatic food chains

7.4 Industrial Emissions

Heavy industries — steel manufacturing, cement production (calcination), chemical synthesis, and petroleum refining — emit vast quantities of CO_2, NO_x, heavy metal particles (lead, mercury, cadmium), dioxins, furans, and VOCs. Cement production alone accounts for ~8% of global CO_2 emissions. Industrial VOCs and NO_x contribute to urban smog formation.

7.5 Vehicular Emissions

With over **1.4 billion motor vehicles worldwide**, vehicular emissions are a dominant factor in urban air quality. Vehicles emit CO_2, NO_x, CO, VOCs, and $PM_{2.5}$ from tailpipes. NO_x from vehicles reacts with atmospheric ozone, contributing to ground-level ozone (smog). Black carbon (diesel soot) is a potent short-lived climate forcer.

> ■ **NOTE:** India has 300+ million registered vehicles. Delhi's Air Quality Index (AQI) regularly exceeds 400 (Severe) in winter due to vehicle emissions + stubble burning + temperature inversion — a combination of multiple atmospheric factors.

7.6 Chlorofluorocarbons (CFCs) & Ozone Depletion

■ What are CFCs?

Chlorofluorocarbons (CFCs) are synthetic compounds (e.g., Freon — CCl_2F_2) historically used as refrigerants in ACs and refrigerators, aerosol spray propellants, and foam packaging agents. They are chemically inert at ground level — but this stability is what makes them so damaging in the stratosphere.

■ Ozone Depletion Mechanism

CFCs rise to the stratosphere (15–50 km altitude) and remain there for **50–100 years**. Intense UV radiation breaks CFC molecules, releasing free chlorine atoms (Cl•). Each chlorine atom catalytically destroys up to **100,000 ozone molecules**:

- Step 1: CF_2Cl_2 + UV → CF_2Cl• + Cl• (CFC breakdown)
- Step 2: Cl• + O_3 → ClO• + O_2 (ozone destroyed)
- Step 3: ClO• + O → Cl• + O_2 (chlorine atom regenerated — cycle repeats)
- Net: O_3 + O → $2O_2$ (ozone permanently lost, UV-B reaches Earth's surface)

■ Effects of Ozone Depletion

- **Increased UV-B Radiation:** The ozone layer shields Earth from harmful UV-B. A thinner layer allows more UV-B to reach the surface
- **Skin Cancer & Cataracts:** UV-B causes melanoma and other skin cancers; damages the eye lens causing cataracts (leading cause of blindness)
- **Immune Suppression:** UV-B suppresses the human immune system, increasing susceptibility to infections
- **Crop Damage:** Excess UV-B reduces crop yields of legumes, rice, and soybeans by damaging DNA and photosynthetic machinery
- **Marine Ecosystem Disruption:** UV-B kills phytoplankton in surface ocean waters — disrupting the base of marine food chains

■ Montreal Protocol (1987)

The Montreal Protocol on Substances that Deplete the Ozone Layer (1987) is widely regarded as the **most successful international environmental treaty**. It phased out production and use of CFCs globally. The ozone layer is slowly recovering and is expected to fully heal by **2060–2070**. However, CFCs already in the stratosphere will continue causing damage for decades.

> ■ **EXAM TIP:** 7-Mark Q: Explain factors responsible for altering the composition of the atmosphere. OR: Describe the CFC ozone depletion mechanism with steps. Include the Montreal Protocol and effects of ozone depletion.

SECTION 8: TECHNIQUES TO CONTROL AIR POLLUTION

8.1 Source Reduction — Prevention at Source

- **Fuel Switching:** Replace coal/diesel with CNG, LPG, natural gas, or renewables. CNG adoption in Delhi's buses dramatically reduced air pollution after mandatory adoption in 2001.
- **Improved Combustion Efficiency:** Advanced burner designs ensure complete burning → less CO and unburned hydrocarbons. Complete combustion: fuel + O_2 → only CO_2 + H_2O.
- **Catalytic Converters:** Three-way catalytic converters in vehicles use Pt/Pd/Rh catalysts: (1) CO → CO_2, (2) HC → CO_2+H_2O, (3) NO_x → N_2. Efficiency ~90%. Requires unleaded fuel.
- **Renewable Energy:** Solar, wind, hydropower generate electricity with zero direct air pollution. India targets 500 GW renewable energy by 2030.
- **BS-VI Emission Norms:** Adopted April 2020 (equivalent to Euro 6). Reduced PM from diesel by 80%, NO_x from diesel by 70% vs BS-IV.

8.2 Particulate Matter Control Devices

■ 1. Cyclone Separator

Principle — Centrifugal Force: Dust-laden air enters tangentially at the top of a cone-shaped chamber. Tangential entry creates a spinning vortex. Heavier particles are flung outward by centrifugal force → contact the wall → spiral downward into a collection hopper at the bottom. Cleaned air exits from a central vortex tube at the top (inner vortex).

- Applications: Cement plants, flour mills, grain elevators, rock crushers
- Efficiency: 70–90% for particles > 10 µm; poor for fine particles ($PM_{2.5}$)
- Advantages: No moving parts, low maintenance, low cost, handles high temperatures
- Limitation: Ineffective for fine particles — often used as pre-cleaner before ESP or bag filter

■ 2. Electrostatic Precipitator (ESP)

Principle — Electric Charge Attraction: Dirty flue gas enters a chamber with high-voltage discharge electrodes (30,000–60,000 volts). Corona discharge ionises the gas, imparting a **negative charge** to fly ash particles. Charged particles migrate toward positively charged collector plates and adhere. Periodic mechanical rapping dislodges collected dust into hoppers below.

- Applications: Thermal power plants (primary fly ash control), cement factories, steel mills
- Efficiency: 99%+ (even for very fine particles and fly ash)

- Advantages: Handles large gas volumes, very high efficiency, low pressure drop
- Limitation: High capital cost; not effective for sticky or electrically resistive particles; high-voltage safety hazard

■ 3. Bag Filter (Fabric Filter)

Principle — Mechanical Filtration: Dust-laden gas passes through woven or felted fabric bags (glass fibre, polyester, PTFE). Particulate matter is trapped on the outer surface of the fabric, forming a dust cake that improves filtration. Bags are periodically cleaned by pulse-jet compressed air blasts — dislodged dust falls into hoppers.

- Applications: Cement, steel, pharmaceutical, food processing, municipal incinerators
- Efficiency: 99%+ for fine dust including PM2.5
- Advantages: Very high efficiency for fine particles; consistent performance; versatile
- Limitation: High pressure drop (energy cost); fabric degradation at high temperatures; bags need periodic replacement

■ 4. Wet Scrubber

Principle — Absorption in Liquid: Polluted gas enters a scrubbing chamber. Water or alkaline liquid (NaOH solution for SO2, lime slurry for acid gases) is sprayed from nozzles, creating a fine mist. Particles collide with liquid droplets and are captured; soluble gases (SO2, HCl, HF) are absorbed into the liquid. Cleaned gas exits from the top; contaminated slurry flows to a treatment system.

- Applications: Chemical plants, acid mist control, SO2 removal (Flue Gas Desulphurisation)
- Efficiency: 90–99% for BOTH particles AND gaseous pollutants simultaneously
- Advantages: Removes particles + gases in one unit; cools hot gases
- Limitation: Creates liquid waste (contaminated slurry); scaling and corrosion; high water use

8.3 Comparison of Control Devices

Device	Principle	Pollutant Removed	Efficiency	Applications
Cyclone Separator	Centrifugal force	Coarse particles (>10 µm)	70–90%	Cement, flour mills
Electrostatic Precipitator	Electric charge	Fine particles, fly ash	99%+	Thermal power plants
Bag Filter	Fabric filtration	Fine particles, PM2.5	99%+	Cement, steel, pharma

Device	Principle	Pollutant Removed	Efficiency	Applications
Wet Scrubber	Liquid absorption	Particles + SO_2, HCl, HF	90–99%	Chemical plants, FGD
Catalytic Converter	Catalytic reaction	CO, NO_x, hydrocarbons	~90%	Automobiles

■ **EXAM TIP:** 7-Mark Q: Describe any two particulate control devices with working principle, diagram description, efficiency, and applications. OR: Compare all four air pollution control devices.

SECTION 9: AMBIENT AIR QUALITY & CONTINUOUS MONITORING

9.1 Definition

Ambient air quality refers to the condition of outdoor air with respect to concentrations of pollutants that may affect human health, ecosystems, and built structures. It is measured and regulated against **National Ambient Air Quality Standards (NAAQS)** — legally prescribed maximum permissible concentrations of specific air pollutants.

9.2 NAAQS — India (CPCB, 2009)

Set by the Central Pollution Control Board (CPCB) under the Air (Prevention and Control of Pollution) Act, 1981.

Pollutant	24-Hour Standard	Annual Standard	Key Health Effect
Sulfur Dioxide (SO_2)	80 µg/m3	50 µg/m3	Respiratory disease, acid rain
Nitrogen Dioxide (NO_2)	80 µg/m3	40 µg/m3	Lung inflammation, smog
PM10 (Coarse particles)	100 µg/m3	60 µg/m3	Respiratory, cardiovascular
PM2.5 (Fine particles)	60 µg/m3	40 µg/m3	Deep lung damage, cancer
Carbon Monoxide (CO)	2 mg/m3 (8-hr avg)	—	Oxygen deprivation
Ozone (O_3)	100 µg/m3 (8-hr avg)	—	Lung damage, crop loss
Lead (Pb)	—	0.5 µg/m3	Neurological damage

■ **NOTE:** WHO 2021 PM2.5 annual limit is 5 µg/m3 vs India's 40 µg/m3 — showing how far Indian cities need to improve. Delhi's annual PM2.5 averages 90–110 µg/m3.

9.3 Continuous Ambient Air Quality Monitoring (CAAQM)

Continuous monitoring uses automated instruments at fixed stations to measure pollutant concentrations **24 hours a day, 365 days a year**, with real-time readings every 15–60 minutes. Far richer data than old manual methods (only 24-hr average, twice a week).

■ Key Components of a CAAQM Station

- **Analyser Instruments:** Specific for each pollutant: UV fluorescence analyser (SO_2), chemiluminescence analyser (NO_x), Beta Attenuation Monitor — BAM (PM2.5/PM10), Non-Dispersive InfraRed — NDIR (CO and CO_2), UV photometric (ozone)
- **Meteorological Sensors:** Wind speed/direction anemometers, temperature/humidity sensors, barometric pressure — meteorological data directly affects pollutant dispersal and concentration
- **Data Logger & SCADA:** Supervisory Control and Data Acquisition systems log measurements every 15 min and transmit wirelessly to a central server for real-time public display
- **Calibration System:** Automatic zero-span calibration using certified reference gas cylinders — ensures instrument accuracy. Typically performed daily/weekly
- **Temperature-Controlled Enclosure + UPS:** Instruments housed in climate-controlled shelters (15–25°C) with Uninterruptible Power Supply — ensures accurate 24/7 operation

9.4 Air Quality Index (AQI)

A standardised index (0–500) translating pollutant concentration data from 8 pollutants (PM10, PM2.5, NO_2, SO_2, CO, O_3, NH_3, Pb) into a single number understandable by the public. The highest sub-index value of any pollutant becomes the overall AQI.

AQI Range	Category	Health Impact	Who is Affected
0 – 50	Good	Minimal impact	None
51 – 100	Satisfactory	Minor discomfort for sensitive people	Very sensitive individuals
101 – 200	Moderate	Breathing discomfort	Asthma, heart disease patients
201 – 300	Poor	Breathing difficulties	General public on prolonged exposure
301 – 400	Very Poor	Serious respiratory effects	Everyone — avoid outdoor activity
401 – 500	Severe/Emergency	Health emergency	Everyone severely affected

9.5 National Programmes

- **NAMP:** National Ambient Air Quality Monitoring Programme — CPCB operates 900+ stations in 344 cities. Monitors SO_2, NO_x, PM10, PM2.5 regularly
- **NCAP 2019:** National Clean Air Programme — India's first national air quality action plan. Targets 20–30% reduction in PM2.5 and PM10 by 2024 vs 2017 baseline in 132 non-attainment cities

■ **EXAM TIP:** 7-Mark Q: Explain Ambient Air Quality Monitoring — NAAQS standards, components of a CAAQM station, AQI with all 6 categories, and national programmes (NAMP, NCAP).

SECTION 10: KYOTO PROTOCOL & CARBON CREDITS

10.1 Kyoto Protocol

The **Kyoto Protocol** is an international treaty adopted in **Kyoto, Japan in December 1997** under the United Nations Framework Convention on Climate Change (UNFCCC). It entered into force on **16 February 2005**. It was the first international agreement to impose **legally binding GHG reduction targets** on developed (Annex I) countries.

■ Greenhouse Gases Covered

Regulated six major GHGs: CO_2, CH_4 (methane), N_2O (nitrous oxide), HFCs (hydrofluorocarbons), PFCs (perfluorocarbons), and SF_6 (sulphur hexafluoride).

■ Commitment Periods

- **First Period (2008–2012):** Developed (Annex I) countries committed to reduce collective GHG emissions by an average of 5.2% below 1990 baseline levels
- **Second Period (2013–2020):** Doha Amendment — more ambitious targets; participated primarily by EU and a few others

■ Principle of CBDR

Common But Differentiated Responsibilities (CBDR): Developed nations (Annex I) historically responsible for most GHG accumulation — bear binding targets. Developing nations (India, China, Brazil) had no binding targets under Kyoto — only voluntary actions.

■ Three Flexibility Mechanisms

- **Emissions Trading (ET):** Countries that reduce emissions below their target can SELL surplus allowances to countries above their targets — 'cap-and-trade'
- **Clean Development Mechanism (CDM):** Developed countries finance emission-reduction projects (renewable energy, reforestation) in developing countries and earn Certified Emission Reductions (CERs). Also transfers clean technology.
- **Joint Implementation (JI):** Two Annex I countries collaborate on emission-reduction projects — investing country earns Emission Reduction Units (ERUs)

■ Limitations & Successor

The **USA refused to ratify** Kyoto in 2001 (then world's largest emitter). Canada withdrew in 2011. The **Paris Agreement (2015)**, signed by 196 countries including USA, India, and China, replaced Kyoto with universal commitments from all nations — though targets are nationally determined rather than legally binding.

10.2 Carbon Credits — Cap-and-Trade System

■ Definition of a Carbon Credit

A **carbon credit** is a tradeable certificate representing the right to emit **one tonne of CO_2** (or CO_2-equivalent of other GHGs). It is the fundamental instrument of emissions trading systems. 1 carbon credit = 1 tonne CO_2.

■ How Cap-and-Trade Works

- Step 1 — CAP set: Government sets maximum total GHG emissions allowed for the year
- Step 2 — Allowances: Total cap divided into allowances (each = 1 tonne CO_2) and distributed to companies
- Step 3 — Under-cap companies: Those that emit LESS than their allocation → SELL surplus credits → earn revenue
- Step 4 — Over-cap companies: Must BUY additional allowances from market OR face large financial penalties
- Step 5 — Cap lowered annually: Overall emissions decline over time while market finds the cheapest reduction paths

■ Types of Carbon Markets

- **Compliance Markets (Mandatory):** EU Emissions Trading System (EU ETS) — 10,000+ installations across 30 countries. India's Carbon Credit Trading Scheme (CCTS, 2023) — first developing country with mandatory domestic carbon market
- **Voluntary Carbon Markets:** Companies voluntarily offset emissions. Standards: Gold Standard, Verified Carbon Standard (VCS). Airlines offering passenger carbon offsetting is an example.

■ Benefits & Criticisms

Aspect	Benefits	Criticisms
Economic	Finds cheapest global emission reduction path	Rich companies 'buy' their way out without actual reduction

Aspect	Benefits	Criticisms
Technology	CDM transfers clean technology to developing nations	CDM projects sometimes had questionable additionality
Flexibility	Countries/companies choose how and where to reduce	Complex MRV systems expensive and can be gamed
Revenue	Funds flow to green projects and reforestation	Carbon leakage — industries relocate to countries with no carbon price

■ **NOTE:** India launched CCTS in 2023 — first developing country with mandatory domestic carbon market. India's NDC: reduce emissions intensity by 45% of 2005 levels by 2030. Net-zero by 2070.

■ **EXAM TIP:** 7-Mark Q: Explain the Kyoto Protocol and Carbon Credits. Include CBDR, three flexibility mechanisms, the cap-and-trade system, and India's CCTS 2023.

SECTION 11: BIOGEOCHEMICAL CYCLES

11.0 Introduction

A **biogeochemical cycle** is the pathway by which a chemical element cycles through the biotic (living organisms) and abiotic (atmosphere, hydrosphere, lithosphere) compartments of the ecosystem. The syllabus covers four cycles: **Carbon, Nitrogen, Sulphur, and Oxygen**.

> ■ **NOTE:** Gaseous cycles (main reservoir = atmosphere): Carbon, Nitrogen, Oxygen. Sedimentary cycle (main reservoir = rocks/soil): Sulphur.

11.1 Carbon Cycle

■ Significance

Carbon is the backbone of all organic molecules and all life. CO_2 is the primary atmospheric form (~0.04% = 420 ppm in 2024, rising rapidly due to fossil fuel combustion). Pre-industrial level was ~280 ppm.

■ Key Processes

- **Photosynthesis (Carbon Fixation):** Plants, algae, and phytoplankton absorb CO_2 and use solar energy to produce glucose: $6CO_2 + 6H_2O + \text{sunlight} \rightarrow C_6H_{12}O_6 + 6O_2$. Ocean phytoplankton account for 50% of all photosynthesis on Earth.
- **Respiration (Carbon Release):** All aerobic organisms break down glucose releasing CO_2: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{ATP}$.
- **Decomposition:** Decomposer bacteria and fungi break down dead organic matter. Aerobic → CO_2. Anaerobic/waterlogged → methane (CH_4).
- **Combustion:** Burning of wood and fossil fuels releases stored carbon as CO_2 — ~36 billion tonnes CO_2 annually. Primary driver of anthropogenic climate change.
- **Ocean Carbon Sink:** Oceans absorb ~25% of anthropogenic CO_2 annually. $CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic acid) → ocean acidification, threatening coral reefs.
- **Geological Sink (Fossil Fuels):** Organic matter buried over millions of years → compressed → forms coal (plants), petroleum, and natural gas. Carbon locked for geological timescales, now released in decades.

■ Human Disturbance

- Fossil fuel combustion: 36 Gt CO_2/year — releasing geological carbon stored over 300+ million years
- Deforestation: removes carbon sinks, releases stored biomass — 10–15% of GHG emissions
- Result: CO_2 risen 280 → 420 ppm → driving 1.1°C global warming, sea level rise, glacial melt

11.2 Nitrogen Cycle

■ Significance

Nitrogen is essential for amino acids (proteins), nucleic acids (DNA, RNA), chlorophyll, and ATP. N_2 makes up 78% of atmosphere but most organisms cannot use it directly — it must first be **'fixed'** into usable forms.

■ 5 Key Steps in Sequence

- **1. Nitrogen Fixation:** N_2 → NH_3 (ammonia). Pathways: (a) Biological — Rhizobium (root nodules of legumes), Azotobacter (free-living in soil), cyanobacteria/BGA — enzyme: NITROGENASE. (b) Atmospheric — lightning converts N_2 → NO_x → HNO_3 → soil. (c) Industrial — Haber-Bosch process: $N_2 + 3H_2 \rightarrow 2NH_3$ at high temperature and pressure (for fertilisers).
- **2. Nitrification:** NH_4^+ in soil → NO_2^- (by Nitrosomonas bacteria) → NO_3^- (by Nitrobacter bacteria). Both steps require aerobic (oxygen-rich) conditions. NO_3^- is the main form absorbed by plant roots.
- **3. Assimilation:** Plants absorb NO_3^- through roots → synthesise amino acids and proteins. Animals obtain nitrogen by consuming plants or other animals.
- **4. Ammonification:** When organisms die, decomposer bacteria and fungi break down nitrogen-containing proteins → release NH_3/NH_4^+ back into soil — making nitrogen available for nitrification again.
- **5. Denitrification:** In anaerobic (waterlogged) soil, denitrifying bacteria (Pseudomonas, Thiobacillus) convert NO_3^- → N_2O → N_2, returning nitrogen to the atmosphere. Completes the cycle.

> ■ **NOTE:** NITROGENASE enzyme catalyses biological nitrogen fixation — present ONLY in prokaryotes. It is irreversibly inhibited by oxygen — which is why Rhizobium root nodules contain LEGHAEMOGLOBIN (pink pigment) to maintain low O_2 levels and protect nitrogenase.

11.3 Sulphur Cycle

■ Significance

Sulphur is essential for life — found in amino acids cysteine and methionine, and vitamins biotin and thiamine. Critical for protein folding through disulfide bonds. The sulphur cycle is primarily **sedimentary** — main reservoir is rocks/soil/ocean sediments.

■ Key Processes

- **Weathering:** Rainfall and erosion break down sulphate rocks — gypsum ($CaSO_4 \cdot 2H_2O$), pyrite (FeS_2) — releasing SO_4^{2-} ions into soil water for plant uptake
- **Assimilation:** Plant roots absorb SO_4^{2-} → incorporate into cysteine and methionine amino acids. Animals obtain sulphur by consuming plants.
- **Decomposition:** Decomposer bacteria break down sulphur-containing proteins → release H_2S ('rotten egg' smell). Thiobacillus bacteria oxidise H_2S → SO_4^{2-}.
- **Volcanic Emissions:** Major natural source of atmospheric SO_2. Mt. Pinatubo (1991): 20 million tonnes SO_2 — cooled global temperatures by ~0.5°C for 2 years (sulphate aerosol cloud reflecting sunlight).
- **Fossil Fuel Combustion:** Burning coal (1–4% sulphur) releases SO_2 — primary cause of acid rain. ~50% of total global SO_2 emissions.
- **Acid Rain Formation:** $SO_2 + H_2O \rightarrow H_2SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$. Acid rain ($pH < 5.6$): acidifies lakes, leaches soil nutrients causing forest dieback, corrodes limestone buildings.

11.4 Oxygen Cycle

■ Significance

O_2 constitutes ~21% of atmosphere, essential for aerobic respiration. Also forms the ozone layer (O_3) in the stratosphere shielding Earth from UV-B. Ocean phytoplankton produce **50–80% of Earth's oxygen**. The oxygen cycle is intimately linked to the carbon cycle through photosynthesis and respiration.

■ Key Processes

- **Photosynthesis (Production):** Primary source of O_2: $6CO_2 + 6H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6O_2$. Phytoplankton produce 50–80% of Earth's O_2.
- **Aerobic Respiration (Consumption):** All aerobic organisms: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{ATP}$. O_2 consumed by global respiration approximately balances O_2 produced by photosynthesis.

- **Combustion (Consumption):** Burning fossil fuels, wood, and biomass consumes O_2 and releases CO_2.
- **Decomposition (Consumption):** Aerobic decomposers consume O_2 to break down dead organic matter → CO_2, H_2O, inorganic nutrients.
- **Rusting / Weathering (Consumption):** Iron + O_2 + H_2O → $Fe(OH)_3$. Slow geological process removes O_2 over millions of years.
- **Photolysis (Minor Production):** UV radiation splits atmospheric water vapour: $2H_2O \rightarrow 4H + O_2$. Ancient source of O_2 before photosynthetic organisms evolved.

■ **NOTE:** Oxygen and Carbon cycles are most interconnected: photosynthesis produces O_2 and consumes CO_2 simultaneously; respiration and combustion do the opposite. Deforestation and ocean pollution disrupt BOTH cycles at once.

■ **EXAM TIP:** 7-Mark Q: Describe the Carbon Cycle OR Nitrogen Cycle with all steps in detail. OR: Describe the Sulphur Cycle with a diagram. How does fossil fuel combustion affect the sulphur cycle and cause acid rain?

SECTION 12: EXAM QUESTIONS — UNIT 1

Part A — 2-Mark Questions

1. Define air pollution. Give two examples each of primary and secondary pollutants.

2. What is photochemical smog? Name the two chemical precursors that form it.

3. Define bioaccumulation and biomagnification. Give one example of each.

4. What is Non-Point Source Pollution? How is it different from Point Source?

5. Name any two sources of SOx and state two health effects.

6. What is the role of Rhizobium in the nitrogen cycle? What enzyme does it use?

7. Define the Kyoto Protocol. When was it adopted and when did it enter into force?

8. What is a carbon credit? What does one carbon credit represent?

9. How do CFCs deplete stratospheric ozone? Name the treaty that banned CFCs.

10. Name the four air pollution control devices and state which is most efficient.

11. What is the Air Quality Index (AQI)? Name all six AQI categories.

12. What is nitrogenase? Why does it require low oxygen conditions?

13. State any two factors responsible for altering the composition of the atmosphere.

14. What is temperature inversion and how does it worsen photochemical smog?

Part B — 7-Mark Questions (OR Type)

■ Q1(a): Explain photochemical smog — define, give its step-by-step formation mechanism, conditions favouring it, all effects (health + environmental), and control measures.

Answer Guide: Define (Los Angeles smog, secondary pollutant) → Formation: 5 steps (NOx + VOCs → UV → photolysis of NO2 → O• → O3 → PAN + smog) → 5 conditions: sunshine, traffic, low wind, temperature inversion, mountains → Health effects: eye/throat irritation, respiratory damage → Environmental: crop damage (ozone phytotoxicity), visibility reduction, rubber cracking → Control: EVs, catalytic converters, BS-VI, CNG, public transport.

■ Q1(b) [OR]: Describe the Nitrogen Cycle with a neat diagram. Explain all 5 steps in detail with the bacteria involved.

Answer Guide: Significance (amino acids, DNA, RNA) → N_2 = 78% but unusable directly → 5 steps: (1) Fixation — Rhizobium, Azotobacter, lightning, Haber-Bosch, enzyme nitrogenase (2) Nitrification — Nitrosomonas (NH_4^+ → NO_2^-), Nitrobacter (NO_2^- → NO_3^-) (3) Assimilation — plants absorb NO_3^- (4) Ammonification — decomposers → NH_3 (5) Denitrification — Pseudomonas → N_2. Mention nitrogenase + leghaemoglobin.

■ Q2(a): Describe all techniques for controlling air pollution. Explain any two particulate control devices with working principle, applications, and efficiency.

Answer Guide: Source reduction: fuel switching, BS-VI, catalytic converter, renewables → Devices: ESP (electric charge, 99%+, power plants — explain corona discharge, collector plates, rapping) + Wet Scrubber (liquid absorption, 90–99%, SO_2 removal — alkaline spray, removes particles AND gases) → Include comparison table of all 4 devices.

■ Q2(b) [OR]: Explain factors responsible for altering the composition of the atmosphere. Include the greenhouse effect and the CFC ozone depletion mechanism.

Answer Guide: Normal composition (N_2 78%, O_2 21%, CO_2 0.04%) → 5 factors: (1) Deforestation — CO_2 sink loss, 10–15% GHG (2) Fossil fuels — 36 Gt CO_2/yr, 280→420 ppm (3) Industrial emissions — metals, VOCs, dioxins (4) Vehicular — NOx, VOCs, black carbon (5) CFCs — chain reaction: Cl• + O_3 → ClO• + O_2 → Cl• (100,000 O_3 per Cl atom) → effects: UV-B, skin cancer → Montreal Protocol 1987. Greenhouse effect: CO_2 traps IR radiation.

■ Q3(a): Explain Indoor Air Pollution — sources, major pollutants, health effects, building syndromes (SBS and BRI), and control measures.

Answer Guide: 90% time indoors, 3.8 million deaths/yr (WHO), 2–5x higher than outdoor → 6 pollutants: CO (gas stoves, 200 ppm→headache, 1600 ppm→death), VOCs (paints, benzene/formaldehyde), PM2.5 (chulha, lung cancer), biological (mold, dust mites), Radon (2nd leading lung cancer cause), Asbestos (mesothelioma) → SBS (no specific illness, resolves when leaving) vs BRI (diagnosable, Legionnaires' disease) → Control: ventilation, clean fuels, low-VOC materials.

■ Q3(b) [OR]: Explain Ambient Air Quality Monitoring — NAAQS standards, components of a CAAQM station, and the AQI system.

Answer Guide: Define ambient air quality → NAAQS by CPCB (2009) under Air Act 1981 — table: SO_2 80 µg/m3, NO_2 80 µg/m3, PM10 100 µg/m3, PM2.5 60 µg/m3, CO 2 mg/m3, O_3 100 µg/m3 → CAAQM: 5 components (UV fluorescence for SO_2, chemiluminescence for NOx, BAM for PM, NDIR for CO, meteorological sensors, data logger, calibration, UPS) → AQI 0–500 in 6 categories → NAMP (900+ stations, 344 cities) → NCAP 2019 (20–30% PM reduction target).

■ Q4(a): Explain the Kyoto Protocol and Carbon Credits. Include CBDR, three flexibility mechanisms, and the cap-and-trade system.

Answer Guide: Kyoto (adopted 1997, Kyoto Japan, in force Feb 2005) → 6 GHGs covered → CBDR principle (Annex I binding, developing nations voluntary) → 2 commitment periods (2008-12: 5.2% reduction; 2013-20: Doha) → 3 mechanisms: ET (cap-and-trade), CDM (developed → developing, earn CERs), JI (Annex I collaboration, earn ERUs) → USA non-ratification 2001 → Paris Agreement 2015 successor → Carbon credit = 1 tonne CO2 → Cap-and-trade 5 steps → India CCTS 2023.

■ Q4(b) [OR]: Describe the Carbon Cycle with a neat diagram. How are human activities disturbing the carbon cycle and causing climate change?

Answer Guide: Significance (backbone of all life, CO2 main atmospheric form) → 6 processes: Photosynthesis (6CO2 + 6H2O → C6H12O6 + O2), Respiration, Decomposition (aerobic=CO2, anaerobic=CH4), Combustion, Ocean sink (25% CO2 absorbed, acidification), Geological (fossil fuels = millions of years stored carbon) → Human disturbance: (1) Fossil fuels 36 Gt CO2/yr (2) Deforestation 10-15% (3) Agriculture (CH4 from rice paddies/cattle, N2O from fertilisers) → CO2 280→420 ppm → 1.1°C warming → glacial melt, sea level rise, ocean acidification.

■ Q5(a): Explain all four biogeochemical cycles — Carbon, Nitrogen, Sulphur, and Oxygen — highlighting the key processes and organisms involved in each.

Answer Guide: Carbon: photosynthesis→respiration→decomposition→combustion→ocean sink→geological. Nitrogen: 5 steps with bacteria (Rhizobium, Nitrosomonas, Nitrobacter, Pseudomonas) + nitrogenase enzyme. Sulphur: sedimentary cycle, weathering→assimilation→decomposition→volcanic→combustion→acid rain (H2SO4). Oxygen: photosynthesis (phytoplankton 50-80%), respiration, combustion, decomposition, rusting — link to carbon cycle.

■ Q5(b) [OR]: Explain the Sulphur Cycle. How does fossil fuel combustion disturb it and cause acid rain? Include control measures.

Answer Guide: Significance (cysteine, methionine amino acids) → Sedimentary cycle, main reservoir rocks/soil → Steps: weathering (CaSO4, FeS2) → SO42- → assimilation → decomposition (H2S) → Thiobacillus (H2S→SO42-) → volcanic emissions (Pinatubo: 20 Mt SO2) → fossil fuel combustion (50% global SO2) → acid rain (SO2+H2O→H2SO4, pH<5.6) → effects: lakes acidified, forest dieback, building corrosion → Control: FGD, low-sulphur fuel, coal desulphurisation.

END OF UNIT 1 NOTES | Environmental Science BHM4T02 | RTMNU NEP 2020 | B.Tech CSE (Data Science)